King Saud University
Muzahimya branch
College of Applied computer science



Subject: Network Protocols

ID Course: cos370

Term: 2026fall

Report title: Write a scenario for 3 TCP messages (piggybacking)

Report No: 1

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m1,A = 81B
m2,A = 5B
m3,A = 33B
ISNA = 3535721075
PORT,A = 110

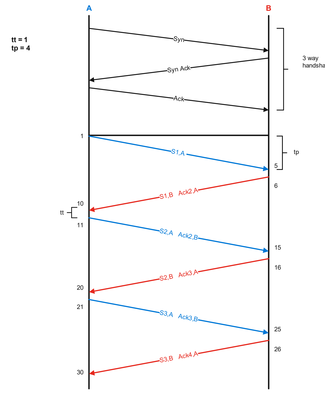
m1,B = 25B
m2,B = 12B
m3,B = 6B
ISNB = 2046860010
PORT,B = 1070

Stream for #A

1-81B	82-86B	87-119B

Stream for #B

1-25B	26-37B	38-43B



TCP Packet Format (Headers & Payloads)

M1,A

110	1070
3535721076	
2046860011	
5B 0101 000000 UAPRSF 011000	64240
0X16E3	0
Option=0B	
Payload=81B	

20B

M1,B

1070	110
2046860011	
3535721157	
5B 0101 000000 UAPRSF 011000	64159
0X75D7	0
Option=0B	
Payload=25B	

20B

M2,A

110	1070
3535721157	
2046860036	
5B 0101 000000 UAPRSF 011000	64215
0X615E	0
Option=0B	
Payload=5B	

20B

M2,B

1070	110
2046860036	
3535721162	
5B 0101 000000 UAPRSF 011000	64154
0XAAB3	0
Option=0B	
Payload=12B	

20B

M3,A

110	1070
3535721162	
2046860048	
5B 0101 000000 UAPRSF 011000	64203
0X25DB	0
Option=0B	
Payload=33B	

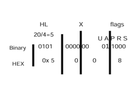
20B

M3,B

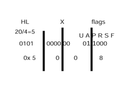
1070	110
2046860048	
3535721195	
5B 0101 000000 UAPRSF 011000	64121
0X4033	0
Option=0B	
Payload=6B	

20B

B13, B14 for S1,A



B13, B14 for others



Convert to Hex

	B1 B2	B3 B4	B5 B6 B7 B8	B9 B10 B11 B12	B13 B14	B15 B16	B17 B18	B19 B20
S1,A	110	1070	3535721076	2046860011	HL + Flag	1460x44	5859	0
	00 6E	04 2E	D2 BE D2 74	7A 00 9A EB	50 18	FA F0	16 E3	00 00
S2,A	B1 B2	B3 B4	B5 B6 B7 B8	B9 B10 B11 B12	B13 B14	B15 B16	B17 B18	B19 B20
	110	1070	3535721076	2046860011	HL + Flag	1460x44	24926	0
	00 6E	04 2E	D2 BE D2 C5	7A 00 9B 04	50 18	FA F0	61 5E	00 00
S3,A	B1 B2	B3 B4	B5 B6 B7 B8	B9 B10 B11 B12	B13 B14	B15 B16	B17 B18	B19 B20
	110	1070	3535721076	2046860011	HL + Flag	1460x44	9691	0
	00 6E	04 2E	D2 BE D2 CA	7A 00 9B 10	50 18	FA F0	25 DB	00 00
S1,B	B1 B2	B3 B4	B5 B6 B7 B8	B9 B10 B11 B12	B13 B14	B15 B16	B17 B18	B19 B20
	1070	110	2046860011	3535721076	HL + Flag	1460x44	30167	0
	04 2E	00 6E	7A 00 9A EB	D2 BE D2 C5	50 18	FA F0	75 D7	00 00
S2,B	B1 B2	B3 B4	B5 B6 B7 B8	B9 B10 B11 B12	B13 B14	B15 B16	B17 B18	B19 B20
	1070	110	2046860011	3535721076	HL + Flag	1460x44	43699	0
	04 2E	00 6E	7A 00 9B 04	D2 BE D2 CA	50 18	FA F0	AA B3	00 00
S3,B	B1 B2	B3 B4	B5 B6 B7 B8	B9 B10 B11 B12	B13 B14	B15 B16	B17 B18	B19 B20
	1070	110	2046860011	3535721076	HL + Flag	1460x44	16435	0
	04 2E	00 6E	7A 00 9B 10	D2 BE D2 EB	50 18	FA F0	40 33	00 00